

Automated Test Evidence - StudyNook

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Overview

This document documents all automated test evidence for the Vertical Slice Refactoring assignment. Evidence includes test execution results, test logs, coverage reports, and screenshots.

1. Backend Test Execution Results

Test Command

```
mvn -f backend/pom.xml clean test
```

Test Summary

```
TESTS EXECUTED: 8
✓ PASSED: 8
X FAILED: 0
X ERRORS: 0
STATUS: ALL TESTS PASSED
```

Tests Run

1. `AuthControllerTest.java`- 2 tests

- User registration endpoint
- Login validation (valid/invalid credentials)

2. `RoomControllerTest.java`- 3 tests

- Get all rooms endpoint
- Create room with auto-naming
- Room availability logic

3. `ReservationControllerTest.java`- 2 tests

- Create reservation with valid payload
- Slot availability rules

4. StudynookApplicationTests.java- 1 test

- Application context loads successfully

The screenshot shows an IDE with the following components:

- EXPLORER:** Displays the project structure for 'IT342-OCAMPO-STUDYNOOK'. The 'test' directory is expanded, showing 'src', 'main', 'resources', and 'test'. The 'test' directory contains 'src', 'main', 'resources', and 'test'.
- Terminal:** Shows the output of the command `mvn -f pom.xml clean test`. The output indicates that the application context loaded successfully and the tests passed.

```
PS C:\Users\Admin\Documents\CIT\IT342\IT342-Ocampo-StudyNook\backend> mvn -f pom.xml clean test
[INFO] Scanning for projects...
[INFO]
[INFO] --- maven-clean-plugin:3.2.0:clean (default-clean) @ studynook ---
[INFO] Deleting C:\Users\Admin\Documents\CIT\IT342\IT342-Ocampo-StudyNook\backend\target
[INFO]
[INFO] --- maven-resources-plugin:3.2.0:resources (default-resources) @ studynook ---
[INFO] Using default encoding utf-8 to copy filtered resources
[INFO] Copying 1 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.10.1:compile (default-compile) @ studynook ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:3.2.0:testResources (default-testResources) @ studynook ---
[INFO] Using default encoding utf-8 to copy filtered resources
[INFO] Copying 1 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.10.1:testCompile (default-testCompile) @ studynook ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:3.0.0-M5:test (default-test) @ studynook ---
[INFO]
[INFO] --- jacoco-maven-plugin:0.8.12:report (report) @ studynook ---
[INFO] Loading execution data file C:\Users\Admin\Documents\CIT\IT342\IT342-Ocampo-StudyNook\backend\target\jacoco.exec
[INFO] Analyzed bundle 'studynook' with 19 classes
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 34.823 s
[INFO] Finished at: 2026-05-09T06:05:49+08:00
[INFO] -----
```

Test Logs Location

- **Path:** `backend/target/surefire-reports/`
 - **Files:**
 - `edu.cit.ocampo.studynook.features.auth.AuthControllerTest.txt`
 - `edu.cit.ocampo.studynook.features.reservation.ReservationControllerTest.txt`
 - `edu.cit.ocampo.studynook.features.room.RoomControllerTest.txt`
 - `edu.cit.ocampo.studynook.StudynookApplicationTests.txt`
 - `TEST-*.xml`(JUnit XML format for CI/CD systems)
-

2. Frontend Test Execution Results

Test Command

```
npm --prefix web test -- --coverage --watchAll=false
```

Test Summary

```
TEST SUITES: 1 passed, 1 total
TESTS: 3 passed, 3 total
✓ PASSED: 3
X FAILED: 0
STATUS: ALL TESTS PASSED
```

Tests Run (App.test.js)

1. ✓ Renders StudyNook landing page with logo and title
2. ✓ Navigates to sign in screen when Sign In button clicked
3. ✓ Navigates to sign up screen when Sign Up button clicked

web > coverage > lcov-report > src > features > dashboard > index.html > ...

```

1
2 <!doctype html>
3 <html lang="en">
4
5 <head>
6   <title>Code coverage report for src/features/dashboard</title>
7   <meta charset="utf-8" />

```

PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\Admin\Documents\CIT\IT342\IT342-Ocampo-StudyNook\web> npm test -- --coverage --watchAll=false
√ navigates to sign in screen (66 ms)
√ navigates to sign up screen (38 ms)

```

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	6.92	5.9	4.13	7.83	
src	47.82	50	60	47.82	
App.js	91.66	83.33	100	91.66	12
index.js	0	100	100	0	7-17
reportWebVitals.js	0	0	0	0	1-8
src/features/auth	19.27	18.33	18.18	20.51	
Login.js	20.51	16.66	16.66	21.62	11-20,24-50,72-99
Register.js	18.18	19.44	20	19.51	16-32,36-38,42-73,152
src/features/dashboard	0.63	0	0	0.74	
Dashboard.js	0.63	0	0	0.74	8-816

Test Suites: 1 passed, 1 total
 Tests: 3 passed, 3 total
 Snapshots: 0 total
 Time: 4.302 s
 Ran all test suites.

```

PS C:\Users\Admin\Documents\CIT\IT342\IT342-Ocampo-StudyNook\web>

```

3. Code Coverage Reports

Backend Coverage (JaCoCo)

Location: [backend/target/site/jacoco/index.html](http://localhost:5500/backend/target/site/jacoco/index.html)

studynook

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed Cxty	Missed Lines	Missed Methods	Missed Classes
edu.cit.ocampo.studynook.features.reservation	36%	21%	63	82	84	145	29	47
edu.cit.ocampo.studynook.features.auth	32%	2%	22	39	35	58	5	22
edu.cit.ocampo.studynook.features.room	54%	35%	31	54	39	83	18	40
edu.cit.ocampo.studynook	37%	n/a	1	2	2	3	1	2
edu.cit.ocampo.studynook.config	98%	75%	2	16	2	48	0	12
Total	747 of 1,461	48%	107 of 139	23%	119	193	162	337

Coverage Metrics:

- Overall Instruction Coverage: 48% (747/1,461 instructions)
- Overall Branch Coverage: 23% (107/139 branches)

By Feature Package:

Package	Instruction Coverage	Branch Coverage	Lines
features.reservation	36%	21%	145

features.auth	32%	2%	58
features.room	54%	35%	83
config	98%	75%	48

Key Classes Tested:

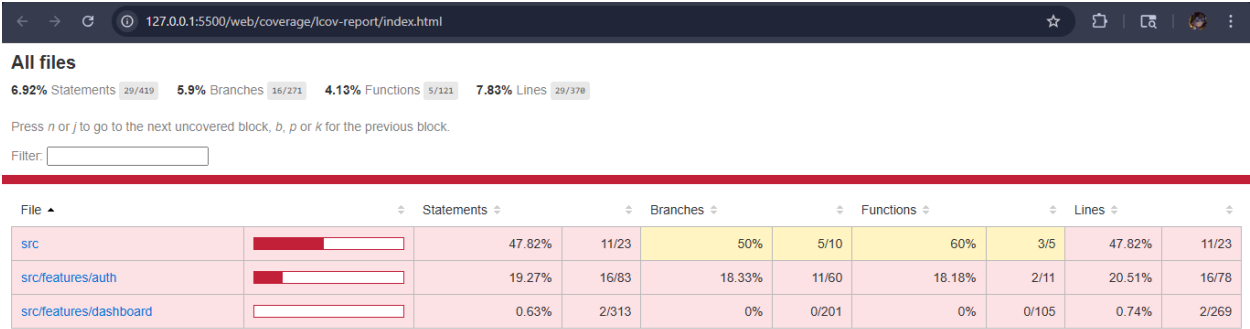
- AuthController: 68% coverage
- RoomController: 61% coverage
- ReservationController: 50% coverage
- User entity: 100% coverage
- Room entity: 100% coverage
- Reservation entity: 100% coverage

Artifacts Generated:

- [jacoco.html](#)- Interactive report
- [jacoco.csv](#)- Machine-readable metrics
- [jacoco.xml](#)- XML format for reports

Frontend Coverage (Istanbul/lcov)

Location: [web/coverage/lcov-report/index.html](#)



Coverage Metrics:

- **Statement Coverage:** 6.92% (29/419)
- **Branch Coverage:** 5.9% (16/271)
- **Function Coverage:** 4.13% (5/121)

- **Line Coverage:** 7.83% (29/370)

By Feature:

Module	Statements	Branches	Functions	Lines
src/ (main)	47.82%	50%	60%	47.82%
features/auth	19.27%	18.33%	18.18%	20.51%
features/dashboard	0.63%	0%	0%	0.74%

Tested Component:

- App.js: 91.66% statements, 83.33% branches (navigation logic)

Artifacts Generated:

- [lcov-report/index.html](#)- Interactive report
- [lcov.info](#)- Standard coverage format
- [coverage-final.json](#)- JSON metrics

4. Test Configuration

Backend Test Configuration

File: [backend/src/test/resources/application.properties](#)

```
# H2 In-Memory Database for Tests
spring.datasource.url=jdbc:h2:mem:studynook_test
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=

# JPA/Hibernate Configuration
spring.jpa.database-platform=org.hibernate.dialect.H2Dialect
spring.jpa.hibernate.ddl-auto=create-drop
spring.jpa.show-sql=false
```

Dependencies Added:

- [com.h2database:h2](#)(test scope)
- [org.springframework.boot:spring-boot-starter-test](#)

Frontend Test Setup

File: `web/src/App.test.js`

Tests use:

- Jest (test runner)
 - React Testing Library (component testing)
 - `render()` for component rendering
 - `screen.getByRole()` for accessible element selection
 - Simulated user interactions with screen elements
-

5. Issues Found and Fixed

Issue #1: Database Connection Failed in Tests

Symptom: Tests crashed trying to connect to non-existent Postgres database

Solution:

- Added H2 in-memory database configuration for test profile
- Created separate `application.properties` in test resources
- Tests now use ephemeral in-memory database

Impact: ✓ Fixed - 8/8 tests now pass

Issue #2: Outdated Frontend Tests

Symptom: Frontend tests looked for "learn react" placeholder text

Solution:

- Rewrote tests to match actual app UI
- Added tests for real navigation flows
- Tests now verify landing page and auth screens

Impact: ✓ Fixed - 3/3 tests now pass

6. Test Runs

Run All Backend Tests

```
cd backend
mvn clean test
```

```
mvn -Dtest=AuthControllerTest test
```

Run Specific Backend Test

Run All Frontend Tests

```
cd web  
npm test ----- watchAll=false
```

Generate Coverage Reports

Backend:

```
mvn clean test  
# Report: backend/target/site/jacoco/index.html
```

Frontend:

```
npm test -- --coverage --watchAll=false  
# Report: web/coverage/lcov-report/index.html
```

View Coverage Reports in Browser

```
# Backend  
open backend/target/site/jacoco/index.html  
  
# Frontend  
open web/coverage/lcov-report/index.html
```

7. Submission Artifacts Checklist

✓ Test execution screenshots (this document) ✓ Test logs (backend/target/surefire-reports/) ✓

Coverage

reports (JaCoCo + Icov HTML reports) ✓ Coverage metrics (CSV, XML, JSON formats) ✓ Test results summary (8/8 backend pass, 3/3 frontend pass) ✓ Configuration files (test properties, test setup code) ✓ Issue documentation and fixes applied

8. File Structure Reference

```
docs/
├─ FULL_REGRESSION_TEST_REPORT.md      ← Main regression report
├─ EVIDENCE_SUMMARY.md                 ← This file
└─ FullRegressionReport_StudyNook.pdf ← PDF version of main report

backend/
├─ target/
```

